XX18/4/CHEMI/HP2/ENG/TZ2/XX

Diploma Programme
Programme du diplôme
Programa del Diploma

Chemistry

Higher level

Paper 2

Thursday 17 May 2018 (afternoon)

Candidate session number

2 hour 15 minutes

Instructions to candidates

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- A calculator is required for this paper.
- A clean copy of the **chemistry data booklet** is required for this paper.
- The maximum mark for this examination paper is **[95 marks]**.

24 pages

Answer all questions. Answers must be written within the answer boxes provided.

1. A student determined the percentage of the active ingredient aluminium hydroxide, $Al(OH)_3$, in a 1.62 g antacid tablet.

The antacid tablet was added to 50.00 cm^3 of 0.120 mol dm^{-3} nitric acid, HNO_3, which was in excess.

(a) Calculate the amount, in mol, of HNO_3. [1]

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(b) Formulate the equation for the reaction of HNO_3 with $Al(OH)_3$. [1]

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(c) The excess nitric acid required 22.40 cm^3 of 0.1050 mol dm^{-3} NaOH for neutralization.

Calculate the amount of excess acid present. [1]

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(d) Calculate the amount of HNO_3 that reacted with $Al(OH)_3$. [1]

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(This question continues on the following page)

(Question 1 continued)

(e) Determine the mass of $Al(OH)_3$ in the antacid tablet. [2]

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(f) Calculate the percentage by mass of aluminium hydroxide in the 1.62 g antacid tablet to three significant figures. [1]

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(g) Outline why repeating quantitative measurements is important. [1]

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2. Graphing is an important tool in the study of rates of chemical reactions.

(a) Sketch a Maxwell–Boltzmann distribution curve for a chemical reaction showing the activation energies with and without a catalyst. [3]

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(b) Excess sulfuric acid is added to lumps of magnesium carbonate. The graph shows the volume of carbon dioxide gas produced over time.

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Volume CO_2 Time

(i) Sketch a curve on the graph to show the volume of gas produced over time if the same mass of **powdered** magnesium carbonate is used instead of lumps. All other conditions remain constant. [1]

(This question continues on the following page)

(Question 2 continued)

(ii) State and explain the effect on the rate of reaction if propanoic acid of the same concentration is used in place of sulfuric acid. [2]

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(c) Outline why pH is more widely used than $[H^+]$ for measuring relative acidity. [1]

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(This question continues on the following page)

(Question 2 continued)

(d) (i) The graph represents the titration of 25.00 cm^3 of 0.100 mol dm^{-3} aqueous propanoic acid with 0.100 mol dm^{-3} aqueous sodium hydroxide.

Volume of 0.100 mol dm^{-3} NaOH (aq) / cm^3

Deduce the **major** species, other than water and sodium ions, present at points A and B during the titration. [2]

A:

B:

(ii) Calculate the pH of 0.100 mol dm^{-3} aqueous propanoic acid. [2]

$K_a = 1.34 \times 10^{-5}$

(This question continues on the following page)

(Question 2 continued)

(iii) Outline, using an equation, why sodium propanoate is basic. [1]

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(iv) Predict whether the pH of an aqueous solution of ammonium chloride will be greater than, equal to or less than 7 at 298 K. [1]

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(e) (i) Formulate the equation for the reaction of sulfur dioxide, SO_2, with water to form sulfurous acid. [1]

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(ii) Formulate the equation for the reaction of one of the acids produced in (e)(i) with zinc carbonate. [1]

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3. The emission spectrum of an element can be used to identify it.

(a) (i) Draw the first four energy levels of a hydrogen atom on the axis, labelling n = 1, 2, 3 and 4. [1]

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Energy

(ii) Draw the lines, on your diagram, that represent the electron transitions to n = 2 in the emission spectrum. [1]

(iii) Hydrogen spectral data give the frequency of $3.28 \times 10^{15}\ s^{-1}$ for its convergence limit. [1]

Calculate the ionization energy, in J, for a single atom of hydrogen using sections 1 and 2 of the data booklet.

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(iv) Calculate the wavelength, in m, for the electron transition corresponding to the frequency in (a)(iii) using section 1 of the data booklet. [1]

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(Question 3 continued)

(b) Elements show trends in their physical properties across the periodic table.

(i) Outline why atomic radius decreases across period 3, sodium to chlorine. [1]

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(ii) Outline why the ionic radius of K^+ is smaller than that of Cl^-. [2]

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(c) (i) Copper is widely used as an electrical conductor. [1]

Draw arrows in the boxes to represent the electronic configuration of copper in the 4s and 3d orbitals.

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(ii) Impure copper can be purified by electrolysis. In the electrolytic cell, impure copper is the anode (positive electrode), pure copper is the cathode (negative electrode) and the electrolyte is copper(II) sulfate solution. [2]

Formulate the half-equation at each electrode.

Anode (positive electrode): **Cathode (negative electrode):**	

(This question continues on the following page)

(Question 3 continued)

(iii) Outline where and in which direction the electrons flow during electrolysis. [1]

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(iv) Deduce any change in the colour of the electrolyte during electrolysis. [1]

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(v) Deduce the gas formed at the anode (positive electrode) when graphite is used in place of copper. [1]

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(d) Explain why transition metals exhibit variable oxidation states in contrast to alkali metals. [2]

Transition metals:

Alkali metals:

4. (a) In acidic solution, chlorate ions, ClO_3^- (aq), oxidize iron(II) ions, Fe^{2+} (aq). [1]

$ClO_3^-\ (aq) + 6H^+\ (aq) + 6e^- \blacksquare\ Cl^-\ (aq) + 3H_2O\ (l)$

$Fe^{2+}\ (aq) \blacksquare\ Fe^{3+}\ (aq) + e^-$

Formulate the equation for the redox reaction.

(b) The change in the free energy for the reaction under standard conditions, ΔG^{o}, is –412 kJ at 298 K. [2]

Determine the value of E^{o}, in V, for the reaction using sections 1 and 2 of the data booklet.

(c) Calculate the standard electrode potential, in V, for the ClO_3^-/Cl^- reduction half-equation using section 24 of the data booklet. [1]

5. Enthalpy changes depend on the number and type of bonds broken and formed.

(a) Hydrogen gas can be formed industrially by the reaction of natural gas with steam. [3
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$CH_4 (g) + H_2O (g) \rightarrow 3H_2 (g) + CO (g)$

Determine the enthalpy change, ΔH, for the reaction, in kJ, using section 11 of the data booklet.

Bond enthalpy for C≡O: 1077 kJ mol^{-1}

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(b) The table lists the standard enthalpies of formation, ΔH^{o}_{f}, for some of the species in the reaction above.

	CH_4 (g)	H_2O (g)	CO (g)	H_2 (g)
ΔH^{o}_{f} / kJ mol^{-1}	–74.6	–241.8	–110.5	

(i) Outline why no value is listed for H_2 (g). [1
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(ii) Determine the value of ΔH^{o}, in kJ, for the reaction using the values in the table. [1
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(Question 5 continued)

(iii) Outline why the value of enthalpy of reaction calculated from bond enthalpies is less accurate. [1]

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(c) The table lists standard entropy, S^o, values.

	CH_4 (g)	H_2O (g)	CO (g)	H_2 (g)
S^o / J K^{-1} mol^{-1}	+186	+189	+198	+131

Calculate the standard entropy change for the reaction, ΔS^o, in J K^{-1}. [1]

$CH_4 (g) + H_2O (g) \rightarrow 3H_2 (g) + CO (g)$

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(d) Calculate the standard free energy change, ΔG^o, in kJ, for the reaction at 298 K using your answer to (b)(ii). [1]

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(e) Determine the temperature, in K, above which the reaction becomes spontaneous. [1]

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6. A mixture of 2.00 mol SO_2 (g), 1.00 mol O_2 (g) and 2.00 mol SO_3 (g) is placed in a 1.00 dm^3 container and allowed to reach equilibrium.

$$2SO_2\ (g) + O_2\ (g)\ ■\ 2SO_3\ (g)$$

(a) Distinguish between the terms reaction quotient, Q, and equilibrium constant, K_c. [1]

(b) The equilibrium constant, K_c, is 0.500 at temperature T. [2]

Deduce, showing your work, the direction of the initial reaction.

(c) (i) Nitrogen oxide is in equilibrium with dinitrogen dioxide. [1]

$$2NO\ (g)\ ■\ N_2O_2\ (g)\quad \Delta H^{o} < 0$$

Deduce, giving a reason, the effect of increasing the temperature on the concentration of N_2O_2.

(This question continues on the following page)

(Question 6 continued)

(ii) A two-step mechanism is proposed for the formation of NO_2 (g) from NO (g) that involves an exothermic equilibrium process. [2]

First step: 2NO (g) ■ N_2O_2 (g) fast
Second step: N_2O_2 (g) + O_2 (g) → $2NO_2$ (g) slow

Deduce the rate expression for the mechanism.

(d) The rate constant for a reaction triples when the temperature is increased from 20.0 °C to 30 °C. [2]

Calculate the activation energy, E_a, in kJ mol^{-1} for the reaction using sections 1 and 2 of the data booklet.

7. Some physical properties of molecular substances result from the different types of forces between their molecules.

(a) (i) Explain why the hydrides of group 16 elements (H_2O, H_2S, H_2Se and H_2Te) are polar molecules. [2]

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(ii) The graph shows the boiling points of the hydrides of group 16 elements. [2]

Explain the increase in the boiling point from H_2S to H_2Te.

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(b) Lewis structures show electron domains and are used to predict molecular geometry. [2]

Deduce the electron domain geometry and the molecular geometry for the PH_2^- ion.

Electron domain geometry:

Molecular geometry:

(This question continues on the following page)

(Question 7 continued)

(c) Resonance structures exist when a molecule can be represented by more than one Lewis structure.

(i) Carbon dioxide can be represented by at least two resonance structures, I and II. [2]

I II

O=C=O :O≡C–O:

(1) (2) (1) (2)

Calculate the formal charge on each oxygen atom in the two structures.

Structure	I	II
O atom labelled (1)	....	....
O atom labelled (2)	....	....

(ii) Deduce, giving a reason, the more likely structure. [1]

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(d) Absorption of UV light in the ozone layer causes the dissociation of oxygen and ozone. [2]

Identify, in terms of bonding, the molecule that requires a longer wavelength to dissociate.

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(e) Carbon and silicon are elements in group 14. [2]

Explain why CO_2 is a gas but SiO_2 is a solid at room temperature.

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8. The structure of an organic molecule can help predict the type of reaction it can undergo.

(a) The Kekulé structure of benzene suggests it should readily undergo addition reactions. [2]

Discuss two pieces of evidence, **one physical** and **one chemical**, which suggest this is not the structure of benzene.

Physical evidence:

Chemical evidence:

(b) (i) Formulate the ionic equation for the oxidation of butan-1-ol to the corresponding aldehyde by acidified dichromate(VI) ions. Use section 24 of the data booklet. [2]

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(ii) The aldehyde can be further oxidized to a carboxylic acid. [2]

Outline how the experimental procedures differ for the synthesis of the aldehyde and the carboxylic acid.

Aldehyde:

Carboxylic acid:

(This question continues on the following page)

(Question 8 continued)

(c) Improvements in instrumentation have made identification of organic compounds routine.

The empirical formula of an unknown compound containing a phenyl group was found to be C_4H_4O. The molecular ion peak in its mass spectrum appears at m/z = 136.

(i) Deduce the molecular formula of the compound. [1]

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(ii) Identify the bonds causing peaks **A** and **B** in the IR spectrum of the unknown compound using section 26 of the data booklet. [1]

A:
B:

(iii) Deduce full structural formulas of **two** possible isomers of the unknown compound, both of which are esters. [2]

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(Question 8 continued)

(iv) Deduce the formula of the unknown compound based on its ^{1}H NMR spectrum using section 27 of the data booklet. [1]

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Organic compounds often have isomers.

9. (a) A straight chain molecule of formula $C_6H_{12}O$ contains a carbonyl group. The compound cannot be oxidized by acidified potassium dichromate(VI) solution.

(i) Deduce the structural formulas of the **two** possible isomers. [2]

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(Question 9 continued)

(iii) Mass spectra **A** and **B** of the two isomers are given. [2]

Explain which spectrum is produced by each compound using section 28 of the data booklet.

A:

B:

(b) A tertiary halogenoalkane with three different alkyl groups, $(R_1R_2R_3)C–X$, undergoes a S_N1 reaction and forms two isomers.

(i) State the type of bond fission that takes place in a S_N1 reaction. [1]

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(ii) State the type of solvent most suitable for the reaction. [1]

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(iii) Draw the structure of the intermediate formed stating its shape. [2]

Shape:

(iv) Suggest, giving a reason, the percentage of each isomer from the S_N1 reaction. [2]

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(Question 9 continued)

(c) Nitrobenzene, $C_6H_5NO_2$, can be converted to phenylamine via a two-stage reaction. [2]

In the first stage, nitrobenzene is reduced with tin in an acidic solution to form an intermediate ion and tin(II) ions. In the second stage, the intermediate ion is converted to phenylamine in the presence of hydroxide ions.

Formulate the equation for each stage of the reaction.

Stage one:

Stage two: